

이차형식

Under $\mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{V})$, $A = A^T$

- $\mathbf{y}^T \mathbf{A} \mathbf{y} \sim \chi^2(\mathbf{r}(\mathbf{A}), \frac{1}{2} \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu}) \Leftrightarrow \mathbf{A} \mathbf{V} = (\mathbf{A} \mathbf{V})^2$
- $\mathbf{y}^T \mathbf{A} \mathbf{y} \perp \mathbf{B} \mathbf{y} \Leftrightarrow \mathbf{B} \mathbf{V} \mathbf{A} = \mathbf{O}$
- $\mathbf{y}^T \mathbf{A} \mathbf{y} \perp \mathbf{y}^T \mathbf{B} \mathbf{y} \Leftrightarrow \mathbf{B} \mathbf{V} \mathbf{A} = \mathbf{O}$ (Under $\mathbf{B}^T = \mathbf{B}$)
- $\mathbb{E}(\mathbf{y}^T \mathbf{A} \mathbf{y}) = \text{tr}(\mathbf{A} \mathbf{V}) + \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu}$ (Not normal, Any A)
- $\text{Cov}(\mathbf{y}, \mathbf{y}^T \mathbf{A} \mathbf{y}) = \mathbf{V}(\mathbf{A} + \mathbf{A}^T) \boldsymbol{\mu}$ (Not normal, Any A)

중회귀

$$\mathbf{y}_{n \times 1} = \mathbf{X}_{n \times p'} \boldsymbol{\beta}_{p' \times 1} + \boldsymbol{\epsilon}_{n \times 1}$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

$$\hat{\mathbf{y}} = \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \mathbf{H} \mathbf{y}$$

$$\mathbf{H}^2 = \mathbf{H}, \mathbf{H}^T = \mathbf{H}, \quad \mathbf{H} \mathbf{X} = \mathbf{X}, \mathbf{H} \mathbf{1} = \mathbf{1}$$

$$\mathbf{e}^T \mathbf{x} = \mathbf{0}, \quad \sum_i \mathbf{e}_i = \sum_i \mathbf{x}_{ij} \mathbf{e}_i = \sum_i \hat{\mathbf{y}}_i \mathbf{e}_i = \mathbf{0}$$

$$\mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu} = \mathbf{X} \boldsymbol{\beta}, \sigma^2 \mathbf{I}_n)$$

$$\text{SST} = \mathbf{y}^T \left(\mathbf{I}_n - \frac{1}{n} \mathbf{J}_n \right) \mathbf{y}$$

$$\frac{\text{SST}}{\sigma^2} \sim \chi^2 \left(n - 1, \frac{1}{2} \boldsymbol{\mu}^T \left(\mathbf{I}_n - \frac{1}{n} \mathbf{J}_n \right) \boldsymbol{\mu} \right)$$

$$\text{SSE} = \mathbf{y}^T (\mathbf{I}_n - \mathbf{H}) \mathbf{y}, \quad \frac{\text{SSE}}{\sigma^2} \sim \chi^2(n - p - 1)$$

$$\text{SSR} = \mathbf{y}^T \left(\mathbf{H} - \frac{1}{n} \mathbf{J}_n \right) \mathbf{y}$$

$$\frac{\text{SSR}}{\sigma^2} \sim \chi^2 \left(p, \frac{1}{2} \boldsymbol{\mu}^T \left(\mathbf{H} - \frac{1}{n} \mathbf{J}_n \right) \boldsymbol{\mu} \right)$$

$$\text{SSE} \perp \text{SSR}, \quad \frac{\text{MSR}}{\text{MSE}} \sim \mathcal{F}(p, n - p - 1, \lambda)$$

$$\hat{\boldsymbol{\beta}} \sim \mathcal{N}_n(\boldsymbol{\beta}, (\mathbf{X}^T \mathbf{X})^{-1} \sigma^2)$$

$$\widehat{\sigma^2} = \text{MSE} = \frac{\text{SSE}}{n - p - 1}$$

Gauss-Markov Theorem

$\mathbb{E}(\boldsymbol{\epsilon}) = \mathbf{0}$, $\text{Var}(\boldsymbol{\epsilon}) = \sigma^2 \mathbf{I}$, $\mathbf{X}$ 가 full rank일 때,

$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ is Best Linear Unbiased Est.

지렛대점

$$\mathbf{X} = \begin{bmatrix} 1, & \mathbf{x}_1^T \\ \vdots & \\ 1, & \mathbf{x}_n^T \end{bmatrix}, \quad \bar{x}_j = \sum_{i=1}^n x_{ij} / n, \quad \boldsymbol{\chi} = \begin{bmatrix} \mathbf{x}_1^T - \bar{\mathbf{x}}^T \\ \vdots \\ \mathbf{x}_n^T - \bar{\mathbf{x}}^T \end{bmatrix}$$

$$h_{ii} = \sum_{j=1}^n h_{ij}^2 = \frac{1}{n} + (\mathbf{x}_i - \bar{\mathbf{x}})^T (\boldsymbol{\chi}^T \boldsymbol{\chi})^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$$

$$\text{M}(i) = (n - 1)(h_{ii} - \frac{1}{n}), \quad \sum_{i=1}^n h_{ii} = p + 1$$

이상점

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H}) \mathbf{y}, \quad \mathbb{E}(\mathbf{e}) = \mathbf{0}, \quad \text{Var}(\mathbf{e}) = \sigma^2 (\mathbf{I}_n - \mathbf{H})$$

$$\text{표준화잔차: } r_i = \frac{e_i}{s \sqrt{1 - h_{ii}}}$$

$$\text{스튜던트화잔차: } r_i^* = \frac{e_i}{s(i) \sqrt{1 - h_{ii}}}$$

$$s^2(i) = \left[(n - p - 1)s^2 - \frac{e_i^2}{1 - h_{ii}} \right] / (n - p - 2)$$

$$\hat{\boldsymbol{\beta}}(i) = \hat{\boldsymbol{\beta}} - \frac{(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_i e_i}{1 - h_{ii}}$$

$$\begin{aligned} t_i &= \frac{y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}(i)}{s(i) \sqrt{1 + \mathbf{x}_i^T [\mathbf{X}(i)^T \mathbf{X}(i)]^{-1} \mathbf{x}_i}} \\ &= r_i \left(\frac{n - p - 2}{n - p - 1 - r_i^2} \right)^{\frac{1}{2}} = r_i^* \sim t(n - p - 2) \end{aligned}$$

영향점

$$\begin{aligned} |\text{DFFITS}(i)| &= \left| \frac{\mathbf{x}_i^T (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}(i))}{s(i) \sqrt{h_{ii}}} \right| \\ &= \left| \left(\frac{h_{ii}}{1 - h_{ii}} \right)^{\frac{1}{2}} r_i^* \right| \geq 2 \left[\frac{p + 1}{n} \right]^{1/2} \end{aligned}$$

$$\begin{aligned} \text{D}(i) &= \frac{(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}(i))^T \mathbf{X}^T \mathbf{X} (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}(i))}{(p + 1)s^2} \\ &= \frac{r_i^2}{p + 1} \frac{h_{ii}}{1 - h_{ii}} \geq F_{\alpha}(p + 1, n - p - 1) \end{aligned}$$

Misspecified Model

$$\mathbf{y} = \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\epsilon} = \mathbf{X}_k \boldsymbol{\beta}_k + \mathbf{X}_r \boldsymbol{\beta}_r + \boldsymbol{\epsilon}$$

$$\tilde{\boldsymbol{\beta}}_k = (\mathbf{X}_k^T \mathbf{X}_k)^{-1} \mathbf{X}_k^T \mathbf{y}$$

$$\mathbb{E}(\tilde{\boldsymbol{\beta}}_k) = \boldsymbol{\beta}_k + (\mathbf{X}_k^T \mathbf{X}_k)^{-1} \mathbf{X}_k^T \mathbf{X}_r \boldsymbol{\beta}_r$$

$$\text{Var}(\tilde{\boldsymbol{\beta}}_k) = \sigma^2 (\mathbf{X}_k^T \mathbf{X}_k)^{-1}$$

$$\widehat{\sigma^2} = \mathbf{y}^T (\mathbf{I}_n - \mathbf{H}_k) \mathbf{y} / (n - k - 1)$$

변수 선택의 기준

$$R_{ak}^2 = 1 - \frac{n - 1}{n - k - 1} (1 - R_k^2) = 1 - \frac{n - 1}{n - k - 1} \frac{\text{SSE}_k}{\text{SST}}$$

$$\frac{1}{\sigma^2} \text{MSE}(\tilde{\mathbf{y}}_k = \mathbf{X}_k \tilde{\boldsymbol{\beta}}_k) = \frac{1}{\sigma^2} \mathbb{E}(\text{SSE}_k) + 2(k + 1) - n$$

$$C_k = \frac{\text{SSE}_k}{\widehat{\sigma^2}} + 2(k + 1) - n$$

일반화 회귀

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{V})$$

$$\mathbf{V} = \mathbf{C} \boldsymbol{\Lambda} \mathbf{C}^T = (\mathbf{C} \boldsymbol{\Lambda}^{1/2})(\mathbf{C} \boldsymbol{\Lambda}^{1/2})^T = \mathbf{T} \mathbf{T}^T$$

$$\mathbf{T}^{-1} \mathbf{y} = \mathbf{T}^{-1} \mathbf{X} \boldsymbol{\beta} + \mathbf{T}^{-1} \boldsymbol{\epsilon}$$

$$\mathbf{Z} = \mathbf{W} \boldsymbol{\beta} + \boldsymbol{\nu}, \quad \boldsymbol{\nu} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$$

$$\hat{\boldsymbol{\beta}}^* = (\mathbf{X}^T \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{V}^{-1} \mathbf{y}$$

$$\mathbb{E}(\hat{\boldsymbol{\beta}}^*) = \boldsymbol{\beta}, \quad \text{Var}(\hat{\boldsymbol{\beta}}^*) = \sigma^2 (\mathbf{X}^T \mathbf{V}^{-1} \mathbf{X})^{-1}$$

$$\text{SSE}(\hat{\boldsymbol{\beta}}^*) = \mathbf{y}^T \mathbf{V}^{-1} \left[\mathbf{I}_n - \mathbf{X} (\mathbf{X}^T \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{V}^{-1} \right] \mathbf{y}$$

$$\mathbb{E}(\text{SSE}(\hat{\boldsymbol{\beta}}^*)) = (n - p - 1) \sigma^2$$

$$\mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$$

$$\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{V} \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1}$$

$$\mathbb{E}(\text{SSE}(\hat{\boldsymbol{\beta}})) = \sigma^2 \text{tr} [\mathbf{V}(\mathbf{I}_n - \mathbf{H})]$$

가변수

범주: $I(x_1 = a)$, 교호: $x_1 I(x_2 = a)$

구간: $y = \beta_0 + \beta_1 x + \beta_2 (x - k) I(x \geq k)$

Variance Inflation Factor

$$\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_p)$$

$$\text{Var}(\hat{\boldsymbol{\beta}}_j) = \sigma^2 \left[(\mathbf{X}^T \mathbf{X})^{-1} \right]_{jj} = \sigma^2 c_{jj}$$

$$c_{jj} = (\mathbf{x}_j^T \mathbf{x}_j - \mathbf{x}_j^T \mathbf{X}_{-j} (\mathbf{X}_{-j}^T \mathbf{X}_{-j})^{-1} \mathbf{X}_{-j}^T \mathbf{x}_j)^{-1}$$

$$R_j^2 = \frac{\mathbf{x}_j^T \mathbf{X}_{-j} (\mathbf{X}_{-j}^T \mathbf{X}_{-j})^{-1} \mathbf{X}_{-j}^T \mathbf{x}_j}{\mathbf{x}_j^T \mathbf{x}_j - (\sum_{i=1}^p x_{ij})^2 / n}$$

$$\text{표준화: } \tilde{\beta}_0 = \bar{y} - \sum_{j=1}^p \tilde{\beta}_j \frac{s_y}{s_j} \bar{x}_j, \quad \tilde{\beta}_j = \frac{1}{s_j} \hat{\beta}_j$$

$$\text{VIF} = c_{jj} = (1 - R_j^2)^{-1}, \quad \text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{1 - R_j^2}$$

Ridge Regression

$$\hat{\boldsymbol{\beta}}(k) = (\mathbf{X}^T \mathbf{X} + k \mathbf{I}_p)^{-1} \mathbf{X}^T \mathbf{y}$$

$$\mathbb{E}(\hat{\boldsymbol{\beta}}(k)) = (\mathbf{X}^T \mathbf{X} + k \mathbf{I}_p)^{-1} \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}$$

$$\text{Var}(\hat{\boldsymbol{\beta}}(k)) = \sigma^2 (\mathbf{X}^T \mathbf{X} + k \mathbf{I}_p)^{-1} \mathbf{X}^T \mathbf{X} (\mathbf{X}^T \mathbf{X} + k \mathbf{I}_p)^{-1}$$

$$\text{MSE}(\hat{\boldsymbol{\beta}}(k)) = k^2 \boldsymbol{\beta}^T (\mathbf{X}^T \mathbf{X} + k \mathbf{I}_p)^{-2} \boldsymbol{\beta}$$

$$+ \sigma^2 \text{tr} \left[(\mathbf{X}^T \mathbf{X} + k \mathbf{I}_p)^{-1} \mathbf{X}^T \mathbf{X} (\mathbf{X}^T \mathbf{X} + k \mathbf{I}_p)^{-1} \right]$$

$$\mathbf{X}^T \mathbf{X} = \mathbf{P} \boldsymbol{\Lambda} \mathbf{P}^T, \quad \mathbf{y} = \mathbf{X} \mathbf{P} \mathbf{P}^T \boldsymbol{\beta} + \boldsymbol{\epsilon} = \mathbf{Z} \boldsymbol{\alpha} + \boldsymbol{\epsilon}$$

$$\hat{\boldsymbol{\alpha}}(k) = (\mathbf{Z}^T \mathbf{Z} + k \mathbf{I}_p)^{-1} \mathbf{Z}^T \mathbf{y} = (\boldsymbol{\Lambda} + k \mathbf{I}_p)^{-1} \boldsymbol{\Lambda} \hat{\boldsymbol{\alpha}}$$

$$\mathbb{E}(\hat{\boldsymbol{\alpha}}(k)) = (\boldsymbol{\Lambda} + k \mathbf{I}_p)^{-1} \boldsymbol{\Lambda} \boldsymbol{\alpha}$$

$$\text{Var}(\hat{\boldsymbol{\alpha}}(k)) = \sigma^2 \boldsymbol{\Lambda} (\boldsymbol{\Lambda} + k \mathbf{I}_p)^{-2}$$

$$\text{MSE}(\hat{\boldsymbol{\alpha}}(k)) = \sigma^2 \sum_{j=1}^p \frac{\lambda_j}{(\lambda_j + k)^2} + \sum_{j=1}^p \frac{k^2 \alpha_j^2}{(\lambda_j + k)^2}$$

$$(\mathbf{X}^T \mathbf{X} = \mathbf{I}) \Rightarrow \hat{k} = p \sigma^2 / \alpha^T \boldsymbol{\alpha}$$

$$\text{PRESS}(k) = \sum_{i=1}^n (y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}^{(i)}(k))^2 = \sum_{i=1}^n \frac{e_i(k)^2}{1 - a_{ii}}$$

$$\hat{\boldsymbol{\beta}}^{(i)}(k) = \hat{\boldsymbol{\beta}}(k) - \frac{(\mathbf{X}^T \mathbf{X} + k \mathbf{I}_p)^{-1} \mathbf{x}_i e_i(k)}{1 - a_{ii}}$$

$$\mathbf{A} = \mathbf{X} (\mathbf{X}^T \mathbf{X} + k \mathbf{I}_p)^{-1} \mathbf{X}^T, \quad e_i(k) = y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}(k)$$

PCR

$$\mathbf{y} = \mathbf{X} \mathbf{P} \mathbf{P}^T \boldsymbol{\beta} + \boldsymbol{\epsilon} \approx \mathbf{X} \mathbf{P}_r \mathbf{P}_r^T \boldsymbol{\beta} + \boldsymbol{\epsilon} = \mathbf{Z} \boldsymbol{\alpha} + \boldsymbol{\epsilon}$$

$$\hat{\boldsymbol{\beta}}^{PCR} = \mathbf{P}_r \hat{\boldsymbol{\alpha}} = \mathbf{P}_r (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{y}$$

$$= \mathbf{P}_r \boldsymbol{\Lambda}_r^{-1} \mathbf{P}_r^T \mathbf{X} \mathbf{y} = \left(\sum_{j=1}^r \frac{\mathbf{p}_j \mathbf{p}_j^T}{\lambda_j} \right) \mathbf{X}^T \mathbf{y}$$

$$\text{bias}(\hat{\boldsymbol{\beta}}^{PCR}) = \left(\sum_{j=r+1}^p \frac{\mathbf{p}_j \mathbf{p}_j^T}{\lambda_j} \right) \boldsymbol{\beta}$$

$$\text{Var}(\hat{\boldsymbol{\beta}}^{PCR}) = \sigma^2 \sum_{j=1}^r \frac{\mathbf{p}_j \mathbf{p}_j^T}{\lambda_j}$$

$$\text{MSE}(\hat{\boldsymbol{\beta}}^{PCR}) = \sigma^2 \sum_{j=1}^r \frac{1}{\lambda_j} + \sum_{j=r+1}^p \alpha_j^2$$

$$\frac{\lambda_1 + \dots + \lambda_r}{\lambda_1 + \dots + \lambda_p} \geq 0.8$$

Logistic Regression

$$\pi(\mathbf{x}) = \mathbb{P}(y = 1 | \mathbf{x}) = \frac{\exp(\mathbf{x}^T \boldsymbol{\beta})}{1 + \exp(\mathbf{x}^T \boldsymbol{\beta})}$$

$$\text{odds} = \frac{\mathbb{P}(Y = 1 | \mathbf{x})}{\mathbb{P}(Y = 0 | \mathbf{x})} = \frac{\pi(\mathbf{x})}{1 - \pi(\mathbf{x})} = \exp(\mathbf{x}^T \boldsymbol{\beta})$$

$$\log \text{odds} = \log \frac{\pi(\mathbf{x})}{1 - \pi(\mathbf{x})} = \mathbf{x}^T \boldsymbol{\beta}$$

$$\text{OR} = \exp(\beta_1)$$

$$\pi(\boldsymbol{\beta}) = (\pi(\mathbf{x}_1^T \boldsymbol{\beta}), \dots, \pi(\mathbf{x}_n^T \boldsymbol{\beta}))^T$$

$$\dot{l}(\boldsymbol{\beta}) = \mathbf{X}^T (\mathbf{y} - \pi(\boldsymbol{\beta})), \quad \ddot{l}(\boldsymbol{\beta}) = -\mathbf{X}^T \mathbf{W}(\boldsymbol{\beta}) \mathbf{X}$$

$$\mathbf{W}(\boldsymbol{\beta}) = \text{diag}(\pi(\mathbf{x}_i^T \boldsymbol{\beta})(1 - \pi(\mathbf{x}_i^T \boldsymbol{\beta})))$$

$$\hat{\boldsymbol{\beta}}^{(t+1)} = \hat{\boldsymbol{\beta}}^{(t)} - \ddot{l}(\hat{\boldsymbol{\beta}}^{(t)})^{-1} \dot{l}(\hat{\boldsymbol{\beta}}^{(t)})$$

$$= [\mathbf{X}^T \mathbf{W}(\hat{\boldsymbol{\beta}}^{(t)}) \mathbf{X}]^{-1} \mathbf{X}^T \mathbf{W}(\hat{\boldsymbol{\beta}}^{(t)}) \mathbf{z}(\hat{\boldsymbol{\beta}}^{(t)})$$

$$\mathbf{z}(\hat{\boldsymbol{\beta}}^{(t)}) = \mathbf{X} \hat{\boldsymbol{\beta}}^{(t)} + \mathbf{W}(\hat{\boldsymbol{\beta}}^{(t)})^{-1} (\mathbf{y} - \pi(\hat{\boldsymbol{\beta}}^{(t)}))$$

	$Y = 0$	$Y = 1$
$X = 0$	n_{00}	n_{01}
$X = 1$	n_{10}	n_{11}

$$\pi_0 = P(Y = 1 | X = 0), \quad \pi_1 = P(Y = 1 | X = 1)$$

$$L(\pi_0, \pi_1) \propto (1 - \pi_0)^{n_{00}} \pi_0^{n_{01}} (1 - \pi_1)^{n_{10}} \pi_1^{n_{11}}$$

$$\hat{\pi}_0 = \frac{n_{01}}{n_{0+}} = \frac{\exp(\hat{\beta}_0)}{1 + \exp(\hat{\beta}_0)}$$

$$\hat{\pi}_1 = \frac{n_{11}}{n_{1+}} = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1)}$$

$$\text{OR} = \frac{\pi_1 / (1 - \pi_1)}{\pi_0 / (1 - \pi_0)}, \quad \beta_1 = \log \text{OR}$$

사례대조군은 OR만 추정 가능, π 대신 q를 사용

Misc.

$$\frac{\partial(\mathbf{c}^T \mathbf{x})}{\partial \mathbf{x}} = \mathbf{c}, \quad \frac{\partial(\mathbf{y}^T \mathbf{A} \mathbf{y})}{\partial \mathbf{y}} = (\mathbf{A} + \mathbf{A}^T) \mathbf{y}$$

$$\mathbf{S} = \mathbf{A}_{22} - \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$$

$$\mathbf{A}^{-1} = \begin{bmatrix} \mathbf{A}_{11}^{-1} + \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{S}^{-1} \mathbf{A}_{11}^{-1} & -\mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{S}^{-1} \\ -\mathbf{S}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{S}^{-1} \end{bmatrix}$$

$$(A + uv^t)^{-1} = A^{-1} - \frac{A^{-1} u v^t A^{-1}}{1 + v^t A^{-1} u}$$

$$|v^t u| \leq \|u\| \|v\|, \quad H X_1 = X_1, \quad H H_1 = H_1$$

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[\|\hat{\theta} - \theta\|_2^2] = \text{tr}(\text{Var}(\hat{\theta})) + \|\mathbb{E}(\hat{\theta}) - \theta\|_2^2$$

$$T = P D P^T, \quad P P^T = I \iff T^T T = T T^T$$

$$T = U D V^T, \quad U U^T = I, \quad V V^T = I$$

$$T^2 = T \Rightarrow \text{rank}(T) = \text{tr}(T)$$